

C - Programming

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11/08/26

CO3 - AT:2

Course Code: CSA0209

Q1) Write a Factorial program by using Recursive Function

Ans Code :

```
#include <stdio.h>
long long factorial(int n) {
    if (n == 0 || n == 1) {
        return 1;
    }
    else {
        return n * factorial(n-1);
    }
}
```

Syntax

```
returntype Factorial (data type
n)
{
    if (condition)
        return value;
    else
        return n * factorial
        (n-1);
}
```

```
int main () {
    int num;
    printf ("Enter a number: \n");
    scanf ("%d", &num);
    if (num < 0) {
        printf ("Factorial of %d is not defined for negative
        numbers. \n");
    }
    else {
        printf ("Factorial of %d = %.lld \n", num, factorial
        (num));
    }
    return 0;
}
```

Sample Output : Enter a number = 5.
Factorial of 5 = 120.

Q2) Write a Binary search program.

Code :

```
#include <stdio.h>

int main() {
    int a[100], n, key, low, high, mid, i;
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }
    scanf("%d", &key);
    low = 0;
    high = n - 1;
    while (low <= high) {
        mid = (low + high) / 2;
        if (a[mid] == key) {
            printf("Element found at position %d", mid + 1);
            return 0;
        }
        else if (a[mid] < key) {
            low = mid + 1;
        }
        else {
            high = mid - 1;
        }
    }
}
```

Input: Sorted Output:
5
10 20 30 40 50
30
output:
Element found at
position: 3